Recombinant Technology: Measurement and Theory

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- However, not much work documenting technological ancestry: which existing technologies a new technology is built from?

This Paper

- Use Wikipedia and the full text of USPTO patents to measure combinations and construct the **genealogy of technologies** ("technological network").

- Document a series of stylized facts about the recombination of existing technologies into new ones

- Provide a simple theory to account for the documented facts.

This Paper: Main Findings

- A typical new technology builds on only small fraction of the technologies available at its birth
  - **sparse recombination**: exploration is limited, not exhaustive
- Some technologies are adopted as ingredients far more often than others
  - **directed search**: some technologies are systematically favored
- Technologies that lead as ingredients in one decade lose ground to newer technologies.
  - search **redirected** toward newer technologies

This Paper: Theory

- A search-based model of recombination with
  - sparse set of candidate ingredients,
  - directed search toward some technologies,
  - search redirected over time toward newer ones

- Key forces: directed search for recombinations and inherited productivity

- **Result**: exponential growth despite limited exploration of potential recombinations
  - neither combinatorics (Weitzman, 1998; Jones, 2023) nor Pareto tails (Kortum, 1997)

Literature

- **New ideas building on existing ideas.**
  - Schumpeter (1911), Romer (1990), Aghion and Howitt (1992)
- **Recombination in patent data.**
  - Youn et al. (2015); Akcigit, Kerr, and Nicholas (2017)
- **The genealogy of technologies.**
  - Kalyani et al. (2025); Hassan, Kalyani, and Lopez Del Valle (2026)
- **Innovation and ideas networks.**
  - Acemoglu, Akcigit, and Kerr (2016); Marx and Fuegi (2022); Liu and Ma (2026); Chen, Liu, and Ma (2026)
- **Models of recombinant growth.**
  - Weitzman (1998); Acemoglu and Azar (2020); Jones (2023)

Data: The Technological Network

Step 1: Identifying Technologies (Nodes)

- Identify technologies using Wikipedia (Kalyani et al., 2025)
  - classify Wikipedia pages into technologies using an LLM ▸ details
  - extract n-grams from the titles and first 10 words of summaries of Wikipedia pages ($n \in \{1, 2\}$)
  - keep an n-gram only if Wikipedia's search returns its own page first
  - group matched n-grams into "technologies" (Wikipedia titles)

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- Use US patent data (1834–2024) to assign a birth year to every technology
  - technology i's birth year (b_i): first year to 10 patents

Step 2: The Genealogy of Technologies (Edges)

- Determine the ingredients (parents) of each technology using patent data

$$\mathcal{P}_i = \{\text{patents mentioning } i\} \qquad \text{(all patents)}$$
$$\mathcal{F}_j = \{\text{patents mentioning } j \text{ filed by } b_j + 5\} \qquad \text{(founding patents)}$$

- Technology i is an ingredient/parent of technology j (denoted by $i \to j$) if at least 10% of j's founding patents mention i:

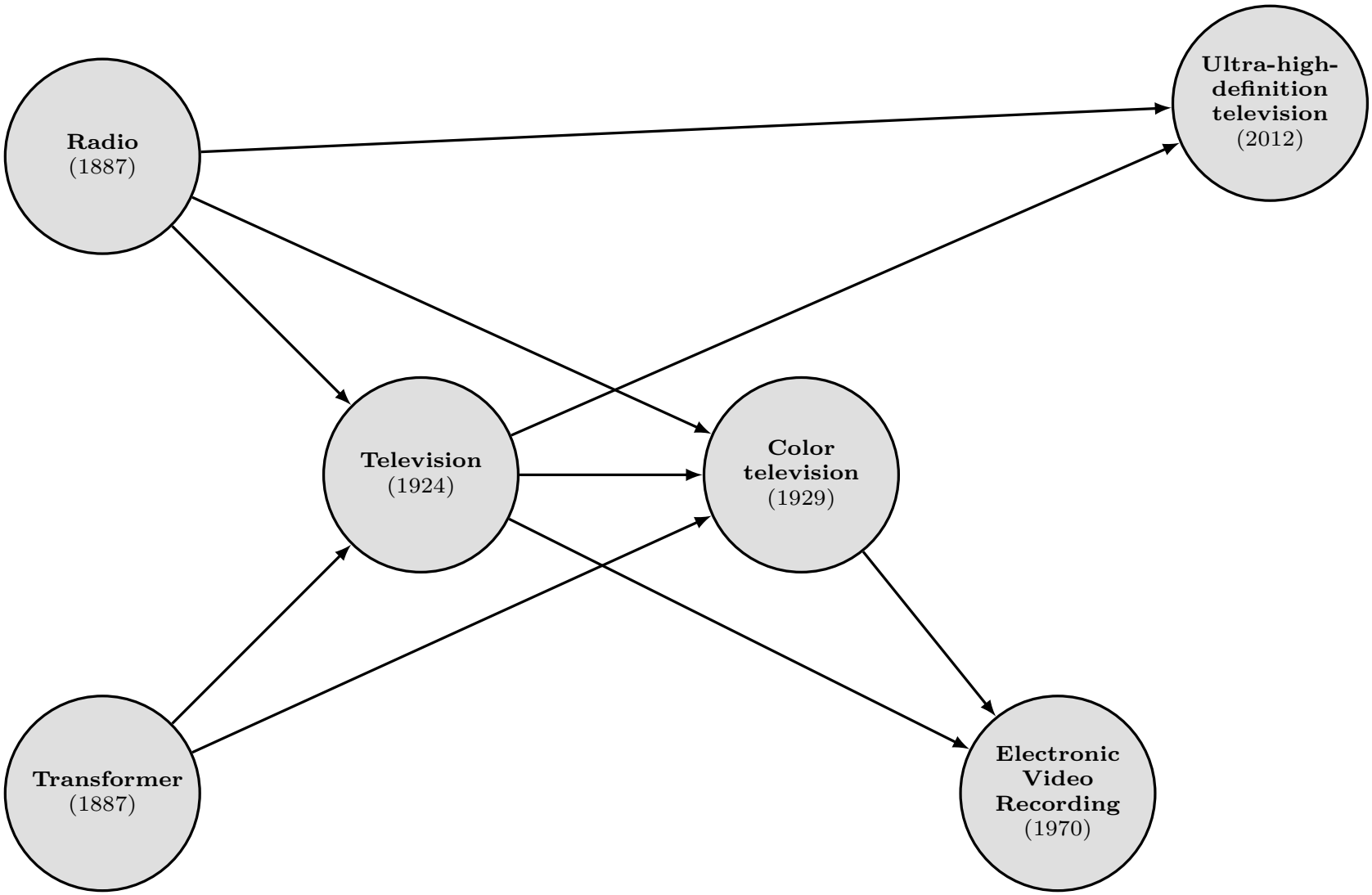
$$\frac{|\mathcal{F}_j \cap \mathcal{P}_i|}{|\mathcal{F}_j|} \geq 10\% \quad \text{and} \quad b_i < b_j$$

The Technological Network

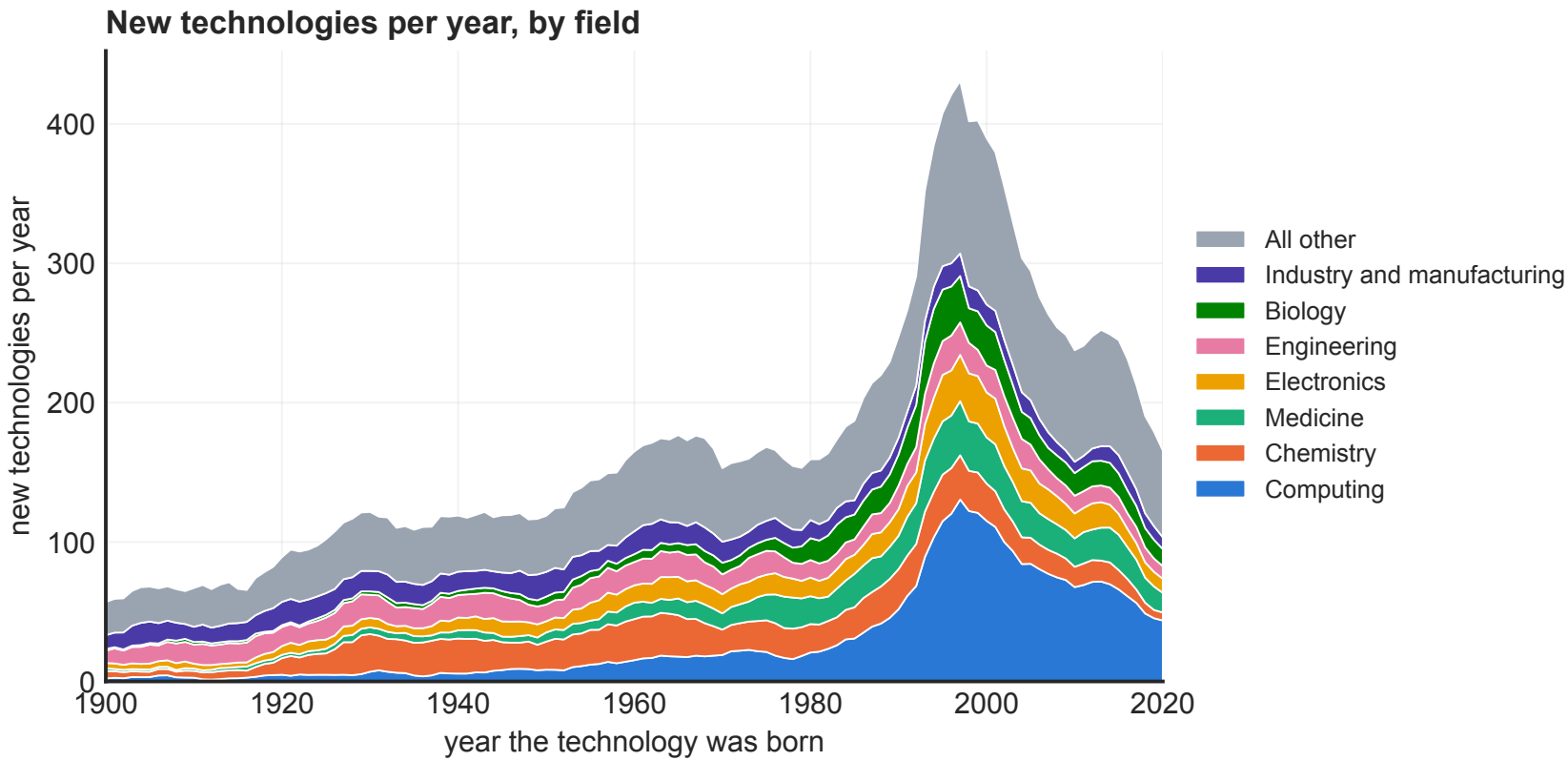
- Technological network: a time series of growing directed acyclic graphs (DAG) with 20,786 technologies, 1834–2024

$n_t = \lvert\{i : b_i \leq t\}\rvert$	technologies born by year t (available stock)
$\mathcal{I}(j) = \{i : i \to j\}$	ingredients of j
$m_j = \lvert\mathcal{I}(j)\rvert$	complexity (in-degree) of j
$\mathcal{C}(i) = \{j : i \to j\}$	children of i
$d_{it} = \lvert\{j \in \mathcal{C}(i) : b_j \leq t\}\rvert$	fertility (out-degree) of i by year t

Technological Network: Granular View



Technological Network: Nodes



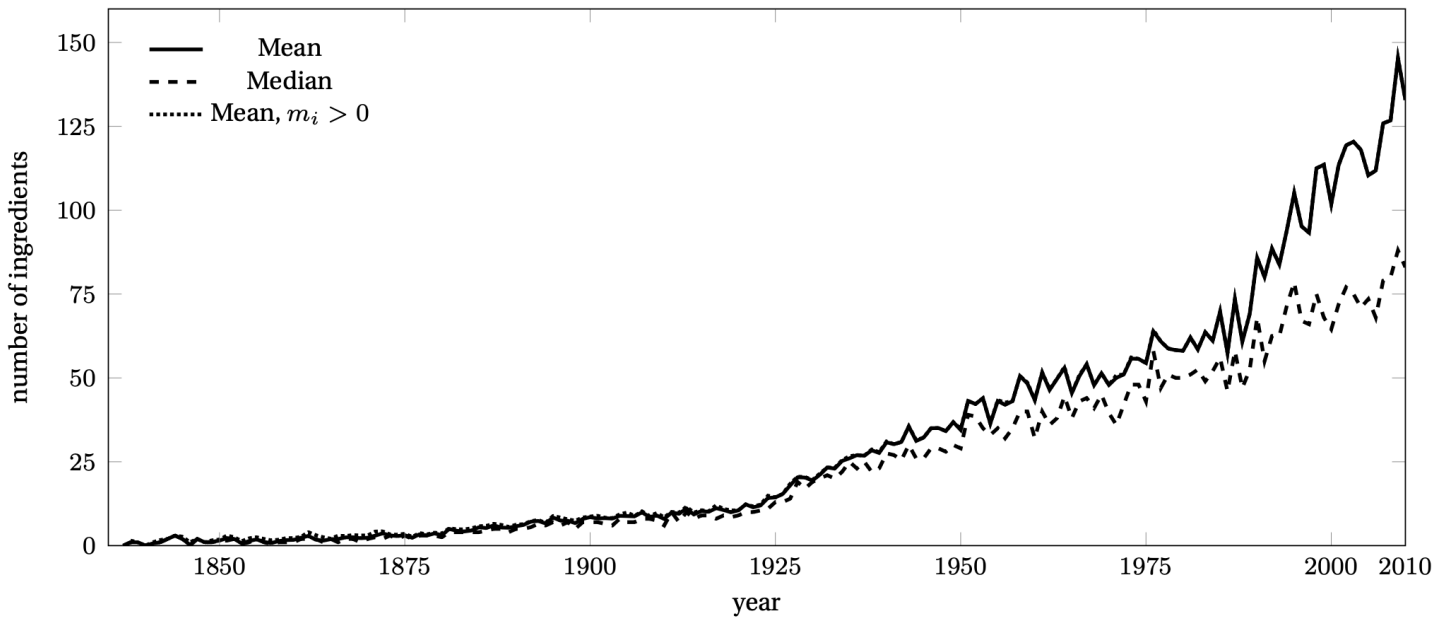
- fields defined based on Wikipedia classification tree

Technological Network: Recombinations

Decade	First recombined	Technology pair	Joint children	Most fertile joint child	Child's children
1900s	1909	Radio; Transformer	112	Capacitor (1926)	127
1910s	1919	Halide; Hydrochloric acid	201	Copolymer (1937)	297
1920s	1925	Ethylene; Halide	188	Copolymer (1937)	297
1930s	1938	Copolymer; Ethylene	253	Acrylonitrile (1939)	158
1940s	1949	Infrared; Isopropyl alcohol	240	Ampicillin (1965)	146
1950s	1950	Chromatography; Dimethyl sulfoxide	225	Ampicillin (1965)	146
1960s	1968	Electronics; Software	472	Internet (1988)	2,570

Stylized Facts

In-Degree (Complexity)

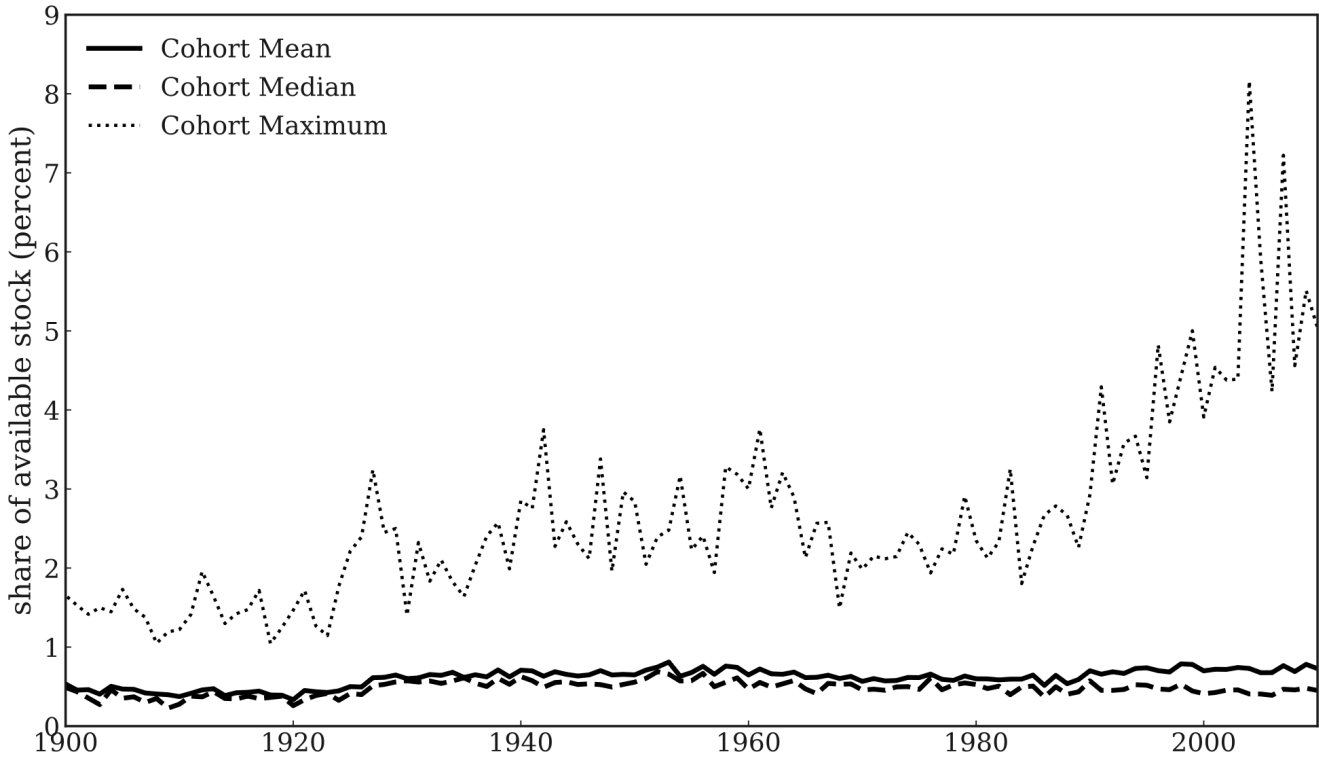


Fact 1. Rising Complexity

The number of ingredients new technologies recombine rises linearly over time.

Normalized In-Degree (Sparsity)

$$s_i = \frac{m_i}{n_{b_i - 1}}$$



Fact 2. Recipe Sparsity

New technologies draw only on a small fraction of available technologies.

In-Degree and Out-Degree Distributions (2010)

	Fertility	Complexity
Observations	17,278	17,278
Mean	58.1	58.1
Standard deviation	245.7	74.8
Median	4.0	35.0
P90	99.0	135.0
P95	236.0	194.0
P99	1,095.4	366.2
Gini coefficient	0.880	0.554

- Out-degree distribution significantly more right-skewed than the in-degree.

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- Out-degree distribution significantly more right-skewed than the in-degree.
- Potential concerns:
  - birth cohorts face different paths of entry and complexity
  - complexity fixed at birth, but fertility accumulates $\rightarrow$ older technologies mechanically have more children

In-Degree and Out-Degree Distributions

- Solution:
  - focus on fertility within the first 20 years after birth, $\Delta_i^{1:20}$
  - normalize by the birth-cohort mean

- For technology i born in year b_i, define

$$\widetilde{m}_i = \frac{m_i}{\frac{1}{|\mathcal{N}_{b_i}|}\sum_{j\in\mathcal{N}_{b_i}} m_j} \quad , \quad \widetilde{\Delta}_i = \frac{\Delta_i^{1:20}}{\frac{1}{|\mathcal{N}_{b_i}|}\sum_{j\in\mathcal{N}_{b_i}} \Delta_j^{1:20}}$$

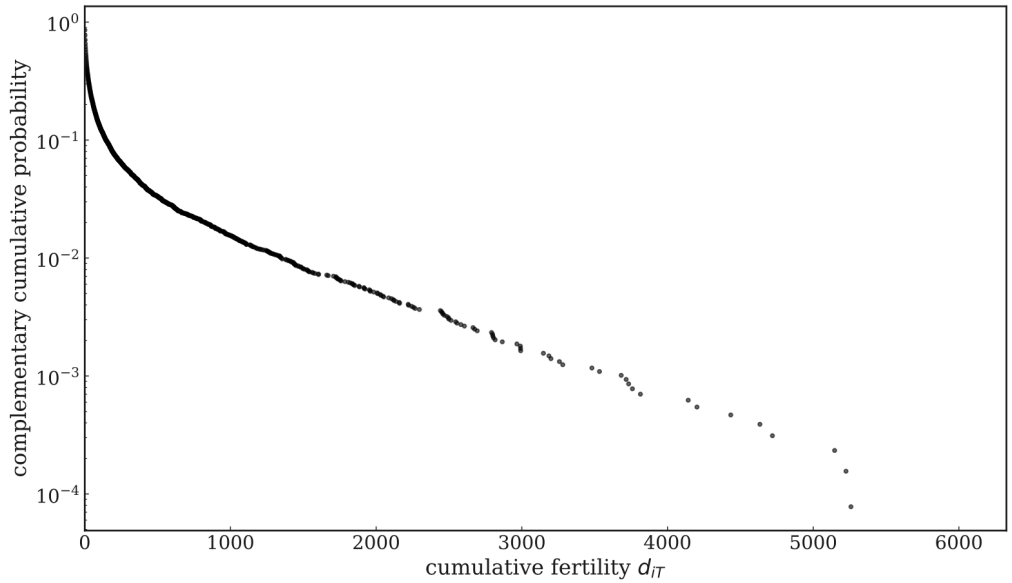
In-Degree and Out-Degree Distributions

	Cohort-normalized measures	
	20-year Fertility	Complexity
Observations	11,043	11,043
Mean	1.0	1.0
Standard deviation	2.791	0.766
Median	0.186	0.818
P90	2.376	2.023
P95	4.413	2.454
P99	12.796	3.442
Gini coefficient	0.795	0.407

Fact 3. Asymmetric Heterogeneity

Within a birth cohort, technologies differ substantially in their complexity, and even more in how often they are subsequently reused as ingredients.

Tail of the Degree Distribution



- Linear fit at the tail on a log-linear scale $\rightarrow$ exponential tail

▶ Formal tests

Fact 4: No Technological Superstars

The upper tail of the fertility distribution is exponential rather than Pareto: the most fertile technologies do not form a separate class of superstars.

Out-Degree Dynamics

- Fertility over the decade starting in t: $\Delta_{it}^{10} = d_{i,t+9} - d_{i,t-1}$
- Fertility leaders, $\mathcal{L}_s$: top 1% most fertile in decade starting in s

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- **Leader Persistence**: fertility of original leaders relative to an equally sized group of current leaders ($\mathcal{H}_{s,t}$)

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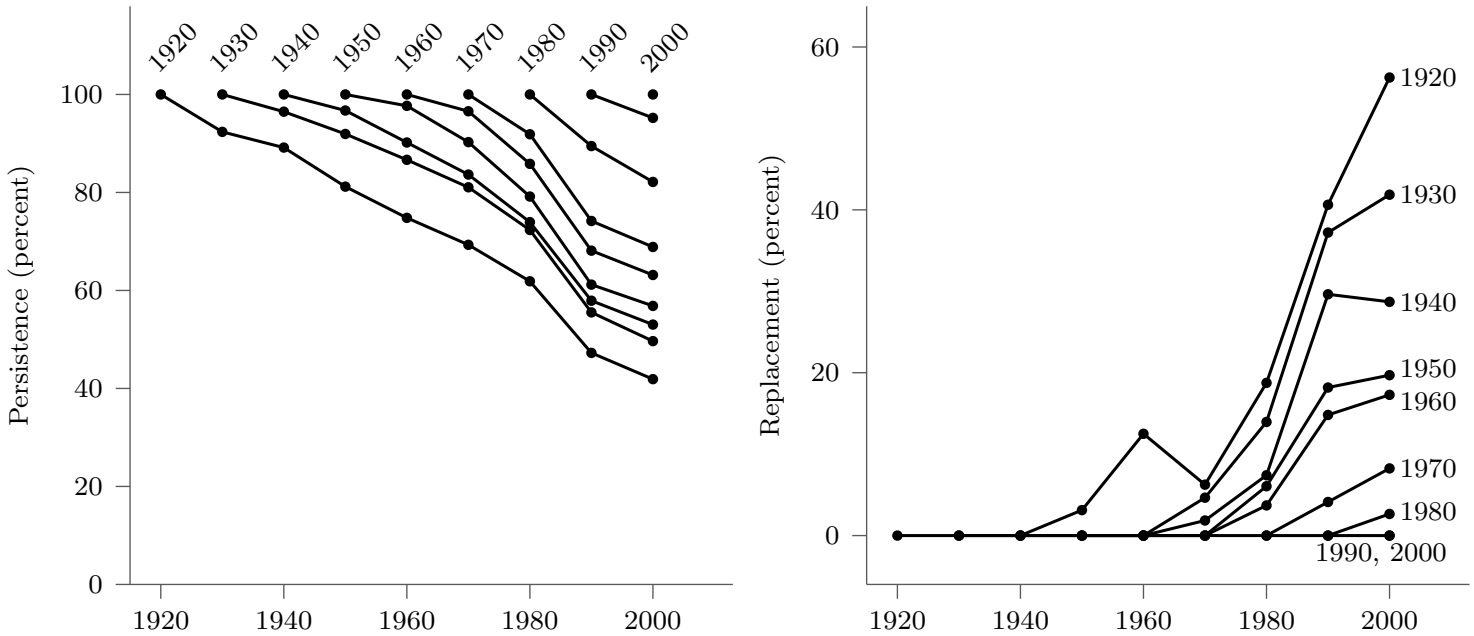
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- **Leader Replacement**: fraction of current leaders born after the original selection decade

$$R_{s,t} = \frac{|\{i \in \mathcal{H}_{s,t} : b_i \geq s + 10\}|}{|\mathcal{L}_s|}.$$

Out-Degree Dynamics



Fact 5. Turnover at the Top

Technologies that lead in fertility in one decade lose ground relative to subsequent leaders, while newer technologies increasingly enter the leading group.

Taking Stock: Search for New Combinations

- New technologies combine few ingredients relative to what is available, even as recipe sizes grow (Facts 1 and 2)
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- **Next.** A model of technological recombination with sparse, directed, and redirected search for new combinations.
  - do sparse, directed, redirected recombination generate sustained growth?

▸ Benchmark models

A Search-Based Model

of Recombinant Growth

Model: Ingredients

1. **Recombination.**
   - New technologies are built by combining existing ones

2. **Search Frictions.**
   - Impossible to examine the full combinatorial space of all possible combinations

3. **Directed Search.**
   - Search is biased toward more productive technologies

4. **Inheritance.**
   - New technologies inherit the productivities of their ingredients: economic growth requires standing on the shoulders of "giants," not "dwarfs"

Model: Environment

- Discrete time economy $t \in \{0, 1, 2, \dots\}$
- n_t technologies at time t with productivities $(A_1, A_2, \dots, A_{n_t})$

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- Inheritance: the productivity of each technology is inherited from its ingredients

$$A_i = \eta_i f\left(\{A_j\}_{j \in \mathcal{I}(i)}\right)$$

  - η_i idiosyncratic productivity shock with bounded support
  - $\mathbb{P}(\eta_i > 1) > 0$: growth is feasible
  - $f(\cdot)$ is strictly increasing in all arguments

Exponential Growth

technological frontier: $A_t^{\max} = \max_{1 \le i \le n_t} A_i$

Theorem

$$\liminf_{t \to \infty} \frac{1}{t} \log A_t^{\max} > 0 \qquad \textit{almost surely}$$

if and only if $\kappa > 0$.

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Theorem

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if and only if $\kappa > 0$.

- The frontier grows exponentially if and only if search is biased toward more productive ingredients
- Without either directed search ($\kappa = 0$) or inheritance the frontier stagnates.

Where Does Growth Come From?

- Kortum (1997)
  - no recombination: exponentially many draws ($k_t = e^{gt}$) with heavy-tailed productivity draws
- Acemoglu and Azar (2020)
  - exploring exponentially many combinations ($k_t = 2^{n_t}$ & $n_t = t$) with heavy-tailed productivity draws
- Jones (2023)
  - exploring doubly exponentially many combinations ($k_t = 2^{n_t}$ & $n_t = e^{\mu t}$) with thin-tailed productivity draws
- This model:
  - linear exploration: $\approx \lambda t$
  - shocks with bounded support
  - growth from productivity inheritance and search bias toward the frontier

Does the Model Replicate the Stylized Facts?

- Assumed:
  - **Fact 2** (sparsity): limited complexity and recombination opportunities
  - **Fact 3** (directed search): search biased toward productive technologies

- Not assumed, so do they follow?
  - **Fact 4**: no superstars
  - **Fact 5**: turnover at the top

Fact 4: No Superstars

Proposition

The cross-sectional distribution of fertility has an exponential tail.

- **Intuition:** no entrenchment
  - since frontier grows exponentially, every technology eventually falls behind
  - search biased toward productive technologies means today's most fertile technologies will not be selected in the future
  - no runaway accumulation of descendants: a thin tail

Fact 5: Turnover at the Top

- Per-period fertility: $\Delta_{it} = d_{it} - d_{i,t-1}$
- Relative fertility: $\chi_{it} = \Delta_{it} / \overline{\Delta}_t$

Proposition

For any technology i born at time t and any $\alpha \geq 1$,

(a) *if $\kappa = 0$, then $\mathbb{E}_t[\chi_{i,t+\alpha}] = 1$ for all α*

(b) *if $\kappa > 0$, then $\mathbb{E}_t[\chi_{i,t+\alpha}] \to 0$ as $\alpha \to \infty$*

- **Intuition:** displacement
  - as the frontier advances, incumbents lose rank, reducing selection prob.
  - $\chi < 1$ can only come from search being redirected toward newer technologies

Testable Predictions

Testable Prediction: Fertility Persistence

Proposition

(a) *if* $\kappa = 0$, *then* $\operatorname{xcov}_t\left(\mathbb{E}_t[\Delta_{it+1}], \mathbb{E}_t[\Delta_{it+\alpha}]\right) = 0$

(b) *if* $\kappa > 0$, *then* $\operatorname{xcov}_t\left(\mathbb{E}_t[\Delta_{it+1}], \mathbb{E}_t[\Delta_{it+\alpha}]\right) > 0$

- While there is turnover across generations, there remain persistent differences within cohorts.
- Intuition: search bias
  - $\kappa = 0$: same selection probability for all cohort members at every age
  - $\kappa > 0$: more productive technologies within a cohort are selected more frequently at all ages

Testable Prediction: Parent-Child Assortativity

Proposition

(a) *if* $\kappa = 0$, *then* $\operatorname{xcov}_{t-1}\left(\mathbb{E}_{t-1}[\Delta_{it}], \mathbb{E}_t[\Delta_{jt+1}]\right) = 0$

(b) *if* $\kappa > 0$, *then* $\operatorname{xcov}_{t-1}\left(\mathbb{E}_{t-1}[\Delta_{it}], \mathbb{E}_t[\Delta_{jt+1}]\right) \geq 0$

- High-fertility parents tend to have high-fertility children
- Intuition: both mechanisms are needed
  - search bias: productive, hence fertile, technologies more likely to be selected
  - inheritance: their children are more productive, hence more fertile

Testable Predictions

	no search bias	search bias, no inheritance	search bias, inheritance
Persistence	$\times$	✓	✓
Assortativity	$\times$	$\times$	✓

Fertility Persistence

- $\Delta_i^{1:10}$ and $\Delta_i^{11:20}$ fertility in the first and second decades of life:

$$\Delta_i^{11:20} = \gamma_{b_i} + \beta\,\Delta_i^{1:10} + \varepsilon_i$$

	Children acquired during ages 11–20
Children acquired during ages 1–10	1.851***
	(0.207)
Birth-cohort fixed effects	Yes
Observations	9,713
Birth cohorts	91
Within R^2	0.669

  - birth-cohort fixed effects: identification from technologies facing the same sequences of entry and aggregate complexity in both windows

Assortativity

- Fertilities in the ten years before and after the child's birth:

$$\begin{aligned} \text{parent:} \qquad & \Delta^{-}_{i,t} = d_{i,t-1} - d_{i,t-11} \\ \text{child:} \qquad & \Delta^{+}_{j} = d_{j,t+10} - d_{j,t} \end{aligned}$$

- Edge-level regression

$$\Delta^{+}_{j} = \alpha_{b_j,\, a_{ij}} + \beta\, \Delta^{-}_{i,b_j} + \varepsilon_{ij}, \qquad a_{ij} = b_j - b_i$$

- child-birth-year $\times$ parent-age-gap fixed effects: compare parents of the same age used by children born in the same year

Assortativity

	Parent–child edges			Children
	Baseline (1)	Recipe-size fixed effects (2)	Equal child weight (3)	Mean parent fertility (4)
Parent's prior fertility	0.0058***	0.0068***	0.0076***	
	(0.0012)	(0.0008)	(0.0010)	
Mean adjusted fertility of parents				0.0707***
				(0.0163)
Child-year × parent-age-gap fixed effects	Yes	Yes	Yes	Pre-adjusted
Child-birth-year fixed effects	—	—	—	Yes
Child recipe-size fixed effects	No	Yes	No	No
Equal total weight per child	No	No	Yes	Yes
Observations	676,890	676,890	676,890	12,610
Children	12,610	12,610	12,610	12,610
Within R^2	0.0012	0.0021	0.0021	0.0202

- Not driven by recipe size: controlling for the child's recipe size or weighting each child equally does not change the results
- Same pattern at the child level: children whose parents had higher (age-adjusted) prior fertility are more fertile themselves

Summary

- **Measurement.**
  - measure recombinations in the data and construct the genealogy of technologies over time ("the technological network") from Wikipedia and patent data, 1834–2024.
- **Findings.**
  - rising complexity, sparse recipes, and heterogeneous fertility, with few superstars and turnover at the top
- **Theory.**
  - limited exploration, search bias, and inherited productivity (i) reproduce these findings and (ii) deliver exponential growth
  - growth needs neither exponentially many draws nor unbounded productivity

Next Steps?

- **Quantitative Exercise.**
  - estimate the model and obtain the implied growth rate
  - time-varying entry λ_t and complexity m_t to account for Fact 1
  - counterfactuals: implications of R&D intensity and the directedness of search for growth?

- **Exploring the data.**
  - **Fields:** within- vs. cross-field recombination?
  - **Breadth:** localized nature of technology?
  - **Productivity:** the key latent variable in the model.

Appendix

Identifying Technologies

1. Wikipedia pages with a section on applications, uses, types, or operation (∼0.5 million pages)

2. Pass through an LLM filter to decide whether the page corresponds to a technology as defined by Schmookler (1966) and Jewkes et al. (1969)

   "a set of specific and applied techniques, products, and processes, as distinct from scientific innovation and from managerial knowledge."

   specifically, the page is classified as a candidate technology if it is **either**

   - a device or system of devices;
   - a process, technique, or set of techniques; **or**
   - a material designed for a specific purpose

   and it describe at least one specific application.

◂ Back

Formal Tests of the Exponential Tail

- Goodness of fit: parametric bootstrap of Clauset, Shalizi, and Newman (2009)
  - $p = 0.45$: cannot reject an exponential tail

- Exponential versus Pareto: log-likelihood ratio (LLR) test
  - LLR $= 6.1$ ($p = 0.024$): favors the exponential over the Pareto

Back

Benchmark Models of Combinatorial Growth

- Example: Jones (2023)
  - n_t different ingredients generate 2^{n_t} possible combinations
  - each possible combination draws productivities independently
  - the combination with maximum productivity is implemented

- **Sparsity**: implemented combinations are highly complex versus 0.5% in the data (Fact 2)

$$\mathbb{E}[s_i] = \frac{1}{2} \qquad \text{and} \qquad \mathrm{sd}(s_i) = \frac{1}{2\sqrt{n_t}}$$

- **Fertility**: independent, anonymous ingredient selection implies little within-cohort dispersion, at most 0.45, compared with 2.89 in the data (Fact 3)

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